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AI/ML Track for HACKaMINeD 2026

EXCEED THE EXPECTED

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PROBLEM STATEMENT

AI-Driven
Solar Inverter Failure Prediction & Intelligence Platform

To predict the likelihood that an solar inverter will experience a shutdown or significant underperformance within a 7–10-day prediction window.

Generate natural language operational summaries and root-cause hypotheses using Generative AI

Business Context

Solar power plants use inverters to convert electricity from solar panels into usable power. If an inverter stops working or performs poorly, the plant produces less electricity. This leads to:

- Loss of energy generation
- Reduced performance ratio (PR)
- Lower uptime
- Financial loss for the company

Business Context

Right now, most solar plants detect inverter problems after they happen using alarms and dashboards. By the time operators notice the issue, some energy has already been lost.

Your solution should improve this by providing two key capabilities:

1. Predictive Visibility

   Use Machine Learning to estimate the risk that an inverter might fail or underperform in the next 7–10 days.

   This helps operators act before energy loss happens.

2. Generative Insight

   Use Generative AI to explain:

   - Why the inverter is at risk
   - What might be causing the issue
   - What actions operators should take

   Instead of just showing numbers, the system should provide clear, easy-to-understand guidance

DETAILS

Machine Learning Component Requirements

Your ML pipeline must demonstrate production-readiness, not just exploratory analysis.

Minimum Requirements

- Binary classification (failure / no failure) or risk score regression within the prediction window.
- Feature engineering from telemetry, computed KPIs, and alarm data.
- Cross-validation and hold-out evaluation with precision, recall, F1, and AUC reported.
- Model explainability: SHAP values or LIME for at least the top 5 contributing features.

Bonus

- Time-series aware splits (e.g., walk-forward validation).
- Anomaly detection as a complementary signal layer.
- Multi-class output: no risk/degradation risk/shutdown risk.

DETAILS

Generative AI Component Requirements

You must integrate an LLM to add a natural language intelligence layer on top of your ML predictions.

Requirements

- Given a prediction output (risk score + feature importances), generate a plain-English summary of why an inverter is at risk and what operational action is recommended. Automated Narrative Generation:
- Allow an operator to ask natural language questions such as 'Which inverters in Block B have elevated risk this week?' and receive grounded, data-backed answers. Retrieval-Augmented Q&A (RAG):
- Document your prompt engineering approach. Show at least two prompt iterations with a rationale for improvements. Prompt Design:

Bonus

- Agentic workflow: the GenAI layer autonomously retrieves inverter data, runs risk assessment, and drafts a maintenance ticket.
- Multi-turn conversational interface with context memory.
- Hallucination guardrails: explicitly show how you prevent the LLM from fabricating inverter data.

DETAILS

Software Engineering & Development Requirements

Backend / API

- Expose your prediction model via a REST API (FastAPI, Flask, or equivalent).
- At minimum: predict endpoint accepting inverter telemetry and returning a risk score + explanation.
- Include a /health endpoint and input validation with informative error messages.
- Containerize with Docker (Dockerfile and docker-compose.yml required).

Frontend / Dashboard

- Build a minimal operational dashboard (React, Streamlit, Dash, or equivalent).
- Must display: per-inverter risk scores, trend visualisations, and the GenAI-generated narrative summary.
- Responsive design is a bonus; correctness and clarity are mandatory.

Code Quality

- Modular, well-commented code with a clear project structure.
- Requirements file (requirements.txt or pyproject.toml) with pinned versions.
- At least 3 unit tests for core ML or API logic.

DETAILS

Submission Requirements

- GitHub repository (public or shared) with complete source code.
- README.md with setup instructions, architecture overview, and design decisions.
- A system architecture diagram (image or draw.io export).
- A 3–5 minute demo video or slide deck demonstrating the working solution end-to-end.
- A short report (max 2 pages) covering: model selection rationale, GenAI design choices, known limitations, and future improvements.

Evaluation Criterias

Component	Requirement	Skills Evaluated	What We Look For
ML Prediction Model	Train a model to predict inverter shutdown or underperformance in a 7–10 day window. Use classification or regression with explainability (SHAP/LIME).	ML, Feature Eng., Model Eval.	Accuracy, explainability, validation rigour
GenAI Copilot / Insight Engine	Build a GenAI layer using an LLM to narrate failure risks, suggest root causes, and answer operator queries in natural language.	GenAI, Prompt Engineering, LLM APIs	Quality of prompts, grounding, and usefulness of outputs
API / Backend Service	Expose predictions via a REST API endpoint. Include input validation, error handling, and a health check endpoint.	Software Engineering, APIs	API design, containerisation, code structure, tests
Dashboard / Frontend	Build a minimal UI showing risk scores, trend charts, and natural language summaries per inverter.	Frontend Dev, Data Viz	Clarity, completeness, usability

Aubergine Datasets

SCAN TO ACCESS

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